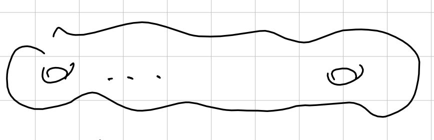
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



($n\geq 1$)
↓
$\pi_1=0$.

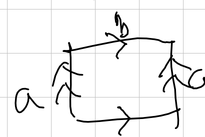
2-dim mfd

$L(p,q)$

$M \bigcirc N$. ∧ ∨

$D^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.

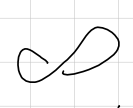
~~$\mathbb{Z}\times\mathbb{Z}$~~

Klein. free product.

ababab b.

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

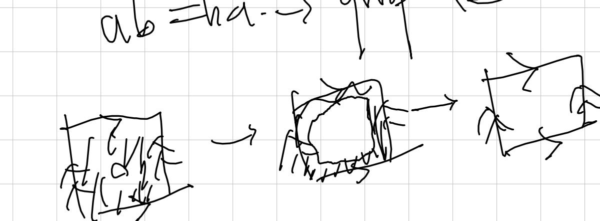


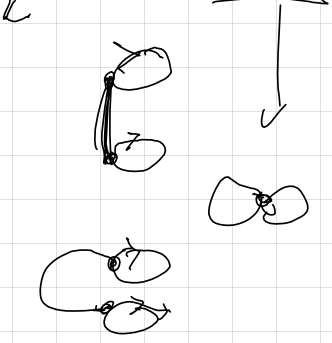
$\langle a,b \mid ab=ba\rangle$

$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

$ab=ba \to$ group +Z.





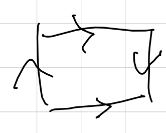
$S^1\vee S^2$

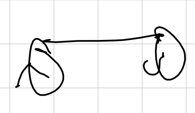
$T: \mathbb{Z}\times\mathbb{Z}$

$\frac{S^1\vee S^1}{\mathbb{Z}*\mathbb{Z}}$

$T_1: \mathbb{Z}*\mathbb{Z}$

$K_1: \mathbb{Z}*\mathbb{Z}$





theorem

S^1
↕
$\mathbb{Z}$

Theorem

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

S^3: $\underline{0}$.
↓
$\mathbb{R}^3$: $\underline{0}$.

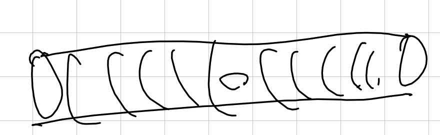
M : n-dim
with $n\geq 3$

then $\pi_1(M) \simeq \pi_1(M\setminus \bullet)$.

$\simeq \pi_1(M\setminus 2pt)$ $\quad$ $\pi_1(S^n)=0$ $\quad n\geq 2$.

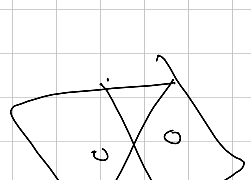
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$
$(n\geq 2)$.

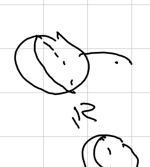
$\simeq \pi_1(M\setminus\{n\})$.

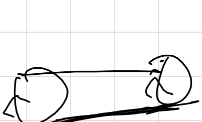


~~tree~~

$S^1\vee S^1\vee\mathbb{R}$.
↓
$\mathbb{Z}*\mathbb{Z}$

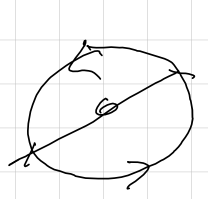


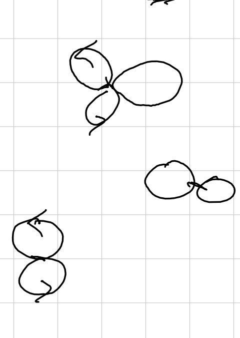




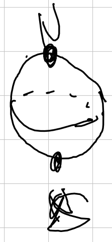
: $S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

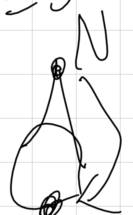


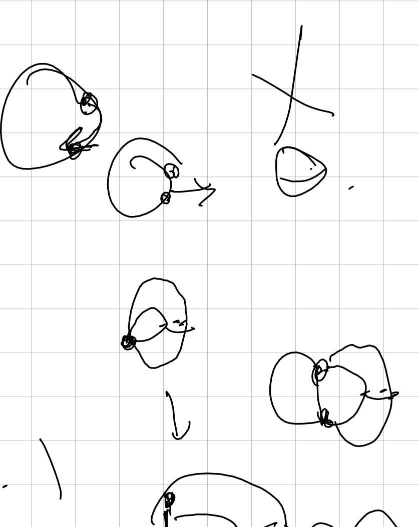


Example.



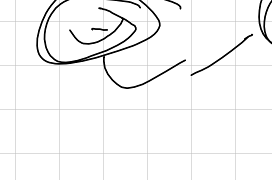
$S^2/N=S$

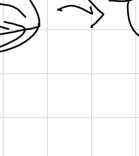




$\to \pi_1(\cdots)$

$S^1\vee S^2$





S^1: 0-cell 1개
1-cell 1개.

S^2: 0-cell 1개
2-cell 1개

$S^1\vee S^1$

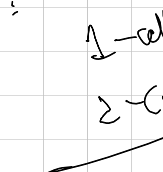
$\mathbb{RP}^2$: 0-cell: 1개
1-cell: 1개
2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba^{-1}\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$ $\quad$ ~~$\mathbb{Z}\times\mathbb{Z}$~~

$\mathbb{RP}^3$: $\mathbb{Z}_2$ $\quad \langle a \mid a^2\rangle$
$\underline{a^2=1}$



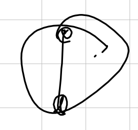
$\mathbb{RP}^3$: ~~0-cell 1개~~
~~1-cell 1개~~
~~2-cell 1개~~
~~3-cell 1개~~

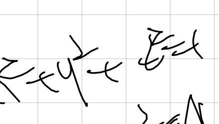
~~$\pi_1(\mathbb{RP}^4)$~~

$\pi_1(\mathbb{RP}^4)=\pi_1(\mathbb{RP}^3) \simeq \mathbb{Z}_2$ ...

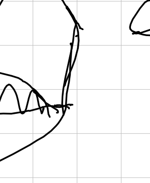
$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$

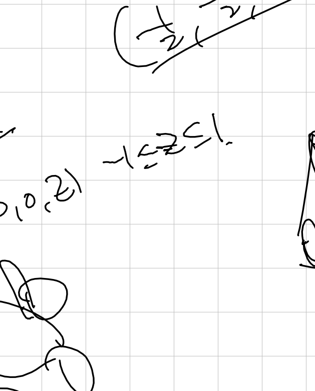
← $x^2+y^2+z^2=1$
∪ $x^2+y^2\leq 1$
∪ $(0,0,z)$ $-1\leq z\leq 1$.

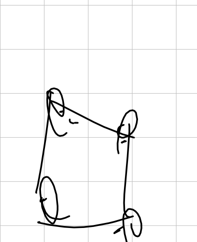


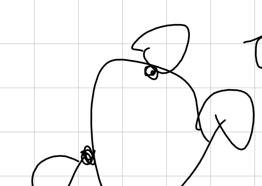


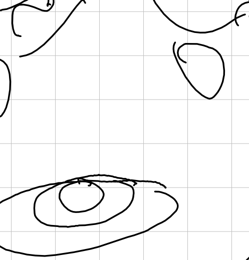
$\mathbb{R}/\mathbb{Z}$

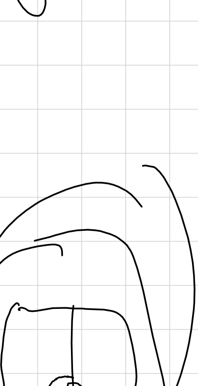


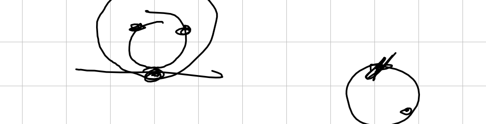










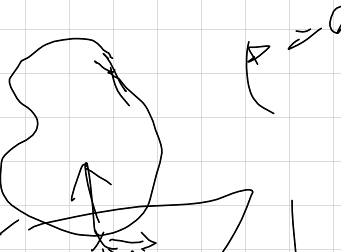


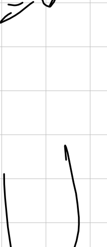
$P=0$

$Z=x^4+y^4$

↓ exp

$K_1=f(4,2)$ ..
$K_2=f(4$ ..





$\frac{1}{P_n+z}$.

$\int \frac{R\,dz}{4}$ $\quad \oint \frac{1}{z^2}dz$ $\quad \frac{4}{2\pi i}$

$2\pi i \cdot X(1^2)$
↓
0